Order Finding And Factoring

Review Learning Objectives

Assignment details

Due
May 19, 11:59 PM IST

Attempts
Unlimited

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Submitted
May 5, 1:26 AM IST

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Next item →

1. What is the order of 7 modulo 15? **1 / 1 point**

- (•) 4
- ( ) 2
- ( ) 1
- ( ) 5

Correct
$7^4 \mod 15 = (49 \mod 15)^2 = 16 \mod 15 = 1$

2. The order of 7 modulo 55 is 20. Select all the correct conclusions from the list below. **3.3333333333333335 / 4 points**

- [x] $7^{10} \mod 55 \neq 1$

  Correct
  Correct.

- [ ] $(7^{10} - 1) \mod 55$ has a common factor with 55 since $(7^{10}+1) \mod 55 \neq 0$
- [ ] $7^{10} \mod 55$ has a common factor with 55.
- [ ] If a is relatively prime to 55 and its order is r then $a^{r/2} - 1 \mod 55$ must have a common factor with 55.
- [x] $(7^{10}+1) \mod 55 \neq 0$, and it has a common factor with 55.

  Correct
  Correct

- [x] $7^{20} \mod 55 = 1$

  Correct

You didn't select all the correct answers

3. Suppose a is relatively prime to a semi-prime number $n = p \times q$ where p, q are prime numbers. **3 / 3 points**

Let r be the order of a modulo n, r be an even number and $a^{r/2} + 1 \mod n \neq 0$.

Select the possible values that $GCD(a^{r/2}+1, n)$ can possibly be.

- [x] q

  Correct

- [ ] $s \times p$ for some $s \geq 2$
- [x] p

  Correct

- [ ] 1
- [ ] n
- [ ] $s \times q$ for some $s \geq 2$

4. Let a, k, n be m bit numbers. What is the complexity of the modular exponentiation **2 / 2 points**

$a^k \mod n$

- Assume that addition, and modulo operation involving m bit numbers can be performed in time $O(m)$ and multiplication + modulo of two m bit numbers involves time $O(m^2)$.

- ( ) $O(m^m)$: m operations of raising one m bit number to the power of another m bit number.
- ( ) $O(m^4)$: m^2 multiplications of m bit numbers.
- ( ) $O(m^2)$: one multiplication of m bit numbers and m modulo operations.
- (•) $O(m^3)$: $2m$ multiplications of m bit numbers and $2m$ modulo operations.

Correct
Correct

5. What can't the "repeated squaring" trick for modular exponentiation used in the notes be applied for order finding? Select all the correct explanations. **2 / 2 points**

- [ ] One can use order finding to make modular exponentiation faster.
- [x] Modular exponentiation is about computing $a^k \mod n$ once for a given k. However, order finding repeatedly needs to compute $a^k \mod n$ for various k until the result is 1.

  Correct
  Correct

- [x] Repeated squaring yields $a \mod n, a^2 \mod n, a^4 \mod n, \cdots, a^{2^k} \mod n$. However the actual order r such that $a^r \mod n$ need not be a power of 2.

  Correct
  Correct